

Security Opportunities and Challenges for Disaggregated Architectures (Invited)

Elisa Bertino
Computer Science Department
Purdue University
West Lafayette, Indiana, USA
bertino@purdue.edu

Imtiaz Karim
Computer Science Department
Purdue University
West Lafayette, Indiana, USA
karim7@purdue.edu

Ashish Kundu
Cisco Research
San Jose, California, USA
ashkundu@cisco.com

Abstract—Disaggregated computer architectures are an interesting paradigm according to which the components of a traditional monolithic server, such as CPU, memory, storage, and networking, are separated into distinct, often independently managed units that communicate over a network. Disaggregation not only offers benefits such as greater flexibility, scalability, and resource optimization but can also improve security. For example, in the context of enterprise routing, it can offer fine-grained control over the network that allows one to deploy security policies, access control rules, and threat detection mechanisms more precisely, ensuring that only authorized traffic flows through the enterprise environment. It makes patch management easier because its modularity allows different components to be patched independently. The same benefits also apply to cellular networks. Disaggregation is a key feature of the Open Radio Access Network (O-RAN) paradigm, whose goal is to make the radio access network intelligent, virtualized, and fully interoperable. However, disaggregation also introduces several unique security risks, such as increased attack surfaces, increased exposure of sensitive data, increased difficulty in tracing data provenance, insecure isolation among different components, and insecure APIs. In addition, well-known security technologies, such as trusted execution environments, may have to be redesigned in the context of disaggregated architectures. In this paper, after an overview of these benefits and concerns, we focus on the research approaches proposed to address some of these concerns for network fabric, O-RAN, and trusted execution environments.

Index Terms—data centers, network fabric, open radio access network, trusted execution environment

I. INTRODUCTION

Disaggregated computer and network architectures decouple hardware components, such as CPUs, memory, storage, and networking resources, into separate modular units. Instead of tightly integrated servers where all components reside in a single machine, disaggregated architectures allow organizations to pool resources dynamically and allocate them as needed. This flexibility is critical for data centers, cloud computing, and high-performance computing, where workloads can vary significantly in resource demands. In the area of database systems, a new type of database systems, called storage disaggregated database systems, has been developed and deployed by major companies, such as Amazon Aurora, Microsoft Socrates,

Google AlloyDB, Alibaba PolarDB, and Huawei Taurus [8]. Network disaggregation follows a similar approach by separating control planes from data planes, enabling software-defined networking (SDN) and more agile network configurations. As noted in [4] “Disaggregated computing is a technology that creates a pool of the component devices of a computer, then combines the necessary devices from the pool to function as a logical computer tailored to ICT service requirements. By extending this concept to the network level across sites, and integrating ICT devices distributed across a network, the entire network can be thought of as a single virtual computer.”

Disaggregation is enabled by technologies such as Compute Express Link (CXL) interconnect— an open interconnect standard designed to enable high-speed, low-latency communication between CPUs, accelerators (e.g., GPUs, FPGAs), and memory. CXL helps address the limitations of traditional PCIe-based communication by allowing memory pooling and coherent memory sharing across heterogeneous computing elements [1], [9]. In addition, progress in the area of photonic communication further enhances the speed of the network fabric, which is critical for disaggregation [4].

Advantages of disaggregation. Disaggregated architectures have several key benefits. One such benefit is resource efficiency. Traditional systems often lead to resource underutilization, as CPUs may be idle while memory is fully used, or vice versa. With disaggregation, workloads are allocated only the resources they require, reducing waste and improving overall performance. In addition, architecture disaggregation supports scalability and flexibility, allowing data centers to upgrade components independently rather than replacing entire servers. This is particularly useful in machine learning (ML) workloads, where GPU-intensive tasks can be separated from general-purpose computing, optimizing cost and performance. Disaggregated architectures also improve resilience and reliability by decoupling compute, memory, storage, and networking resources into independent modular components. This separation allows systems to dynamically reallocate resources in response to failures, load fluctuations, or security threats, minimizing downtime and improving overall system availability. By decoupling critical resources, disaggregated systems maintain high availability and minimize disruptions, even in the presence of hardware or software failures.

The work reported in this paper has been partially funded by NSF under grants 2112471 and 2229876.

Network disaggregation, particularly in SDN and disaggregated cellular networks, such as open radio access network (O-RAN), enables better traffic management. By separating network functions from proprietary hardware, organizations can deploy software-driven solutions that automate threat detection and mitigation while reducing vendor lock-in. Disaggregated networks also enhance programmability, allowing for more adaptive security measures that can quickly respond to emerging cyber threats and allow for automatic rerouting of data in case of link or switch failures, reducing downtime.

Security risks and challenges. Despite its advantages, disaggregation introduces new security risks. The modular nature of these architectures expands the attack surface, as each disaggregated component represents a potential point of failure or intrusion. Data-in-transit security becomes critical, as information must move across different physical and logical components rather than within a single machine. Attackers may exploit these interconnects to conduct man-in-the-middle (MITM) attacks, data leakage, or latency-based exploits. In addition, compromised components can have cascading effects, where an attacker gains access to one module (e.g., memory or network fabric) and escalates privileges to control other critical resources. Another concern is supply chain security. Since disaggregated architectures often involve multi-vendor environments, ensuring consistent security policies and trusted hardware across all components becomes challenging. Malicious firmware or compromised modules introduced at the manufacturing or supply chain level could lead to hardware backdoors or covert data exfiltration risks. Furthermore, security mechanisms such as trusted execution environments (TEE) and memory encryption must be extended across disaggregated systems to prevent unauthorized access.

In what follows, we focus on selected subsystems, namely, the network fabric and the O-RAN. We also briefly discuss the challenges in using trusted execution environments (TEEs) in the context of disaggregated architectures and a possible approach to address these challenges.

II. SECURITY OF THE NETWORK FABRIC

While current network fabrics enable high-performance disaggregated computing, they also introduce new security risks, some of which are well known and affect conventional systems and networks. These risks include: (1) Memory snooping and side-channel attacks as shared memory pools may expose data to side-channel attacks if isolation mechanisms, such as memory, are not properly hardened. (2) Compromised fabric controllers, as attacks on fabric switches and interconnect controllers can compromise resource allocation. (3) Malicious workload injection as attackers could leverage the remote direct memory access (RDMA) mechanism or CXL.mem to inject rogue workloads into shared compute pools. (4) Interconnect hijacking, as attackers could exploit vulnerabilities in high-speed interconnects to reroute or disrupt data flows.

Securing the network fabric requires properly deploying multiple security techniques, such as: (1) memory encryption (such as Intel total memory encryption [6]); (2) memory access

control, such as the access control mechanism introduced by CXL 3.0 [2], restricting memory sharing based on identity and permissions; (3) authenticated and encrypted data transfers, which in turn requires using TLS for moving data across the network fabric and the use of protocols for authenticated key exchange (AKE); (4) zero-trust network fabric security, which applies zero-trust principles by continuously verifying device, user, and workload identities before granting access to shared resources [3]; (5) ML-driven anomaly detection for detecting unusual memory access patterns, unauthorized resource allocations, or lateral movement within the network fabric. Those are just examples, and many other security techniques are relevant for securing the network fabric. The main challenge is minimizing security overhead on the network fabric.

III. O-RAN

The disaggregation of O-RAN has substantially expanded the threat landscape of attacks. The reasons include: (1) Increased remote access. The open and multi-vendor architecture relies on a diverse range of suppliers, developers, and service providers with varied levels of access to components. As a result, the number of personnel potentially accessing sensitive information grows, increasing the risk of misconfiguration or compromised authentication process. (2) The distributed nature of edge servers, often located in accessible public spaces such as sidewalks, rooftops, or building basements, makes on-site physical access an increasingly viable attack entry point. (3) The multi-vendor dependency can create software supply chain risk from untrusted hardware and software vendors.

Recent works have shown that the fronthaul- a vital part of O-RAN disaggregated communications is amenable to various attacks. In O-RAN, the radio access network is split between the Control Unit (CU), Radio Unit (RU), and the Distributed Unit (DU). The fronthaul connects the DU and RU. There are four topologies based on the location of the clocked source and the switched network for the synchronization between the RU and the DU. The synchronization depends on the Precision Time Protocol (PTP). The standards do not mandate any security for the fronthaul, assuming the attacks to be infeasible and requiring heavy sophistication. However, the newly discovered fronthaul attacks are not only feasible but also do not require overly sophisticated hardware and can directly manipulate fronthaul traffic [5], [10]. Furthermore, these attacks are feasible on real-world commercial-grade O-RAN networks, highlighting the impact and prevalence of the issues. In the following, we discuss some of these highly impactful attacks on the O-RAN fronthaul.

Amplification and modification attacks [10]. The high-level idea behind these attacks is to force the devices - aka User Equipments (UEs) to generate signaling messages at a high rate towards the CU. Consequently, the CU can become unresponsive, negatively impacting the users in a broad region beyond the cells directly under attack. In the attack, a radio event called the Radio Link Failure (RLF), which occurs when the radio link between the UE and DU is lost, is utilized. When this happens, the UE seeks to recover the signal and sets a

timer in the process (typically a few hundred milliseconds). The UE triggers a process called cell *re-selection* and triggers the establishment of a new connection. This cell re-selection generates a long sequence of messages to/from the CU, and since a single cell can have hundreds of UEs, forcing all of them to perform cell re-selection causes a signaling storm to the CU. The attack steps are fairly straightforward; an attacker periodically steers the fronthaul packets of cells from one RU to another by swapping the MAC addresses. This action makes the RUs transmit the signal of different cells every few seconds, making the UEs experience an RLF and periodically trigger the cell re-selection process. This attack can also be triggered by giving the UEs the “illusion” that the signal quality of their attached cell goes below some threshold compared to a neighboring cell—modifying the I/Q sample scaling and replacement payloads. This event triggers the handover process, which creates a signal storm by forcing a re-configuration in the context of the UE.

The signal modification attacks modify radio signals and break the connection and/or performance of UEs in a cell region. The most straightforward way to conduct this attack is to corrupt regular packets by replacing I/Q samples with invalid user payload (with random or all zero). From the UE perspective, this would result in decoding failures and packet drops during signal processing. The other attack strategies involve changing the Synchronization Signal Block (SSB) and modifying the Physical Random Access Channel (PRACH). The SSB block includes the necessary information for a 5G UE to attach a cell and connect to the network. Any failure in the receipt or decoding of SSB elements renders UEs unable to attach. To carry out this attack, an adversary can selectively alter specific bits within the SSB. For instance, they can introduce erroneous configurations. The attacks render new UEs unable to establish a connection and cause already connected UEs to detach from the cell. In a similar vein, an attacker can manipulate PRACH packets and prevent new UEs from attaching to a cell while permitting already connected UEs from attaching to continue using the cell without interruption.

Spoofing and replay attacks [5]. This second group of attacks utilize the PTP synchronization packets to carry out the attack. The first type of attack targets PTP nodes by sending counterfeit Announce messages to alter the clock selection process—which is essential for synchronization. Initially, the attacker monitors traffic to gather critical information and the configuration. With the data, the attacker crafts Announce messages designed to have the same attributes but high priority. These crafted messages are sent periodically to maintain the attack. Disrupting the synchronization process, the attacker creates a gradual drift between the clocks and creates a complete outage on the base station. Other attack options involve sniffing and replaying the time synchronization message (Sync and FollowUp) in the PTP protocol and transmitting them after a delay. Similar to the previous attack, this attack also creates a huge offset on the network synchronization and disrupts the base station communications.

Defense and future work. Attacks on the O-RAN fronthaul

are just the tip of the iceberg. With disaggregation in the whole network and the expanded threat surface, securing O-RAN requires delving into multiple security challenges and techniques. One of the potential solutions is to design advanced monitoring and detection approaches. In this domain, previous works have developed a state-of-the-art transformed-based ML detection mechanism to detect some of the fronthaul attacks [5]. For example, the TIMESAFE approach by Groen et al. [5] achieves 97.99% accuracy in detecting attacks on deployed DUs. The ML model based on transformer processes input sequences with fixed lengths, varied during training and evaluation to include varying length of packets based on historical data. The model deploys a custom Weighted Binary Cross-Entropy Loss function to address class imbalance and unequal error costs. Not only providing a solution, this work also shows why sophisticated ML-based approaches are needed to deal with security threats. General rules or heuristic-based solutions are inadequate to detect these attacks where the signature of the attacks can be easily reshaped. For a concrete evaluation, the work by Groen et al. shows that in a real-world commercial grade O-RAN compliant 5G network, the heuristic-based approach has only a 50.95% accuracy and 17.80% recall compared to the transformed-based ML solution with an accuracy of 97.99% and 98.33%, respectively.

Whereas a monitoring and detection mechanism is the first step, for a comprehensive and secure ecosystem, substantial research is needed. Some of these research directions are as follows. First and foremost, the standards need to update and revise the threat landscape. According to the O-RAN Security Work Group, the standardization body responsible for formulating security specifications for the fronthaul, the lack of protection in the fronthaul is acceptable because of the low likelihood of attacks, requiring sophistication on the part of the attackers, and the low severity of the potential attacks. However, the above-mentioned research on real-world commercial O-RAN networks highlights that these threat assumptions do not hold anymore. An extensive revision of the O-RAN specifications is needed to detect potential threat areas, such as the fronthaul. Second, new integrity and authentication schemes need to be devised to secure these disaggregated components with stringent time and memory requirements. External security mechanisms, including MACSec (layer 2) and IPsec (layer 3), might be potential solutions. However, they have some major shortcomings. For instance, the O-RAN Alliance specifications generally do not permit IP transport. Using MACsec poses challenges as software implementations create virtual interfaces that disrupt hardware timestamping, reducing accuracy significantly. This increases costs and delays and reduces the virtualization central to O-RAN. On the whole this requires substantial research and the design of new integrity and authentication schemes that take these requirements into account. Third, zero-trust O-RAN security, which applies zero-trust principles by continuously verifying all the components of the networks, can be a game changer. Last, ML-driven anomaly detection to detect and prevent unauthorized access needs to be deployed to minimize security

risks with such stringent memory and time requirements.

IV. TRUSTED EXECUTION ENVIRONMENTS (TEEs)

TEEs are an important security building block in that they ensure code integrity and provide strong isolation from untrusted parties. However, their use in disaggregated architectures poses several issues [7]: (1) *Loss of physical isolation*. Traditional TEEs assume that a secure enclave is confined within a single physical processor. In disaggregated architectures, computation and memory are often distributed across different physical machines, making it difficult to maintain the strong isolation guarantees that TEEs rely on. (2) *Difficulty in securing communication across disaggregated resources*. TEEs depend on local memory encryption and integrity checks to protect sensitive data within the CPU boundary. However, in a disaggregated setting, data frequently travels across high-speed interconnects, such as CXL, exposing data to potential interception, manipulation, or side-channel attacks during transit. (3) *Attestation complexity in a multi-component environment*. In conventional TEEs, remote attestation allows users to verify the integrity of an enclave before executing trusted workloads. However, in a disaggregated system, multiple independent components must be attested to simultaneously and in a way that guarantees their trustworthiness as a collective system. This makes coordinated attestation across heterogeneous resources a major challenge. (4) *Dynamic resource allocation risks*. Unlike monolithic systems where TEEs operate on fixed hardware, disaggregated architectures dynamically allocate resources on demand. This means that a workload may be assigned memory or compute units that were previously used by other tenants, raising concerns about residual data leakage, side-channel attacks, or compromised components being integrated into secure execution environments. (5) *Heterogeneity and compatibility issues*. TEEs are often designed for specific processor architectures. However, disaggregated architectures incorporate diverse heterogeneous hardware, including ARM processors, GPUs, FPGAs, and specialized accelerators. Ensuring that all such components can participate in a unified, trusted execution environment is a major technical challenge.

To address these challenges, Koshiba et al. [7] introduce a hardware/OS co-design for building virtual TEEs (vTEEs) able to provide an isolated execution environment across multiple disaggregated components. The approach relies on the combination of several techniques: (1) *Secure context migration and isolation*. vTEE extends traditional TEE capabilities by allowing the secure migration of execution contexts between disaggregated nodes without exposing sensitive data to the untrusted infrastructure. To secure data, vTEE uses cryptographic mechanisms ensuring that state transfers between different computing resources are properly protected. (2) *Remote attestation for disaggregated components*. vTEE uses a hierarchical attestation mechanism, where each component (e.g., CPU, memory, accelerators) provides verifiable proof of integrity before being incorporated into a secure execution environment. This approach ensures that workloads are executed only on

trusted hardware, even in distributed settings. (3) *Confidential and authenticated communication*. A major security issue in disaggregated architectures is that TEEs traditionally rely on isolated memory within a monolithic system, but in disaggregated architecture, data must be transmitted across a network fabric. vTEEs thus use secure channels (e.g., TLS, MAC-based authentication, or hardware-supported encryption techniques) to prevent unauthorized access or modification of sensitive data while it moves across the network fabric. (4) *Memory and compute integrity protection*. vTEE leverages encrypted memory access and integrity verification techniques to ensure that only authorized compute units can access specific memory regions. This prevents data leakage and manipulation even when using shared or distributed memory pools. However, while the vTEE approach significantly improves security in disaggregated architectures, several challenges remain including: performance overhead, arising from memory encryption and securing remote communications introduce latency; scalability to support large-scale disaggregated infrastructures; and hardware support as the effectiveness of TEE depends on hardware security primitives, which need broader adoption across diverse architectures.

V. CONCLUDING REMARKS

Protecting disaggregated systems is a formidable task, which requires using many different security techniques. The main challenge is to minimize the security overhead and the cost of security management. Future developments in the area of quantum computing will further complicate security by requiring the deployment of quantum resistant cryptography.

REFERENCES

- [1] About CXL, <https://computeexpresslink.org/about-cxl/>, accessed on March 20, 2025.
- [2] Compute Express Link (CLX). Specification, 3.1, 2022.
- [3] C. Katsis, E. Bertino, “ZT-SDN: an ML-powered Zero-trust Architecture for Software-defined Networks”, *ACM Transactions on Privacy and Security*, Vol. 28, No. 2, Article 23, February 2025.
- [4] Fujitsu, “Photonic Disaggregated Computing to Support ICT Infrastructure in the 6G Era”, White Paper, 2025.
- [5] J. Groen, S. Di Valerio, I. Karim, D. Villa, Y. Zhang, L. Bonati, M. Polese, S. D’Oro, T. Melodia, E. Bertino, F. Cuomo, K. Chowdhury, “TIMESAFE: Timing Interruption Monitoring and Security Assessment for Fronthaul Environments”, <https://arxiv.org/pdf/2412.13049>, 2024.
- [6] Intel, “Intel® Total Memory Encryption”, White Paper, <https://www.intel.com/content/www/us/en/architecture-and-technology/vpro/hardware-shield/total-memory-encryption.html>, accessed on March 21, 2015.
- [7] A. Koshiba, F. Gust, J. Pritzi, A. Vahldiek-Oberwagner, N. Santos, P. Bhatotia, “Trusted Heterogeneous Disaggregated Architectures”, *Proceedings of the 14th ACM SIGOPS Asia-Pacific Workshop on Systems, APSys 2023*, Seoul, Republic of Korea, August 24-25, 2023.
- [8] X. Pang, J. Wang, “Understanding the Performance Implications of the Design Principles in Storage-disaggregated Databases”, *Proceedings of the ACM on Management of Data*, Volume 22(3): 180 (2024).
- [9] D. D. Sharma, R. Blankenship, D. Berger, “An Introduction to the Compute Express Link (CXL) Interconnect”, *ACM Computing Surveys*, 56, 11, Article 290, July 2024.
- [10] J. Xing, S. Yoo, X. Foukas, D. Kim, M.K. Reiter, “On the Criticality of Integrity Protection in 5G Fronthaul Networks”, *Proceedings of the 33rd USENIX Security Symposium*, August 14–16, 2024, Philadelphia, PA, USA.